

On conditional expectations and sigma algebras

In this assignment, we explore conditional expectations and their computation for the random walk stochastic process. We will gradually develop all the ingredients needed to compute conditional expectations given the information contained in some σ -algebra. A general sample space will be denoted Ω , with more specific notation like Ω_N reserved for sample spaces involving coin tosses. Note that the ideas and results discussed below are guaranteed to apply only to situations where the sample space has a finite number of elements, and where the random variables are assumed to take a finite number of values. The sample space and random variables of the random walk satisfy these conditions. For other kinds of sample spaces and random variables, one must carefully reconsider each result and check their validity.

The random walk stochastic process $\mathbb{M}_n$, $0 \leq n \leq N$, is defined as follows. Define $\mathbb{M}_0 = 0$. Then, we define the independent random variables $\mathbb{X}_i$ which depend only on the i^{th} coin toss, specified by

$$\mathbb{X}_i(\omega_i) = \begin{cases} +1 & \omega_i = H , \\ -1 & \omega_i = T . \end{cases} \quad (1)$$

The random variable $\mathbb{M}_n$ is then defined as the sum of the first n $\mathbb{X}_i$:

$$\mathbb{M}_n(\omega_1 \cdots \omega_n) = \sum_{i=1}^n \mathbb{X}_i(\omega_i) , \quad \text{for } n = 1, \dots, N . \quad (2)$$

We also need the definition of an event. Simply said, an event is a subset of the sample space. Usually, an event is a particular subset defined by a rule or a collection of rules that the elements of the subset must satisfy. For instance, the event $\mathbb{M}_1 = 1$ is a subset of Ω_N containing sequences of coin tosses which start with a H .

Given a collection of events $A_1, A_2, \dots, A_k$ in a sample space Ω , the σ -algebra generated by the events is the collection of subsets obtained by taking all possible unions, intersections and complements of $A_1, A_2, \dots, A_k$. This σ -algebra is written as $\sigma(A_1, \dots, A_k)$.

Note: This σ -algebra also turns out to be the smallest σ -algebra containing $A_1, \dots, A_k$. That is, among all σ -algebras containing the events $A_1, \dots, A_k$, the σ -algebra $\sigma(A_1, \dots, A_k)$ is the one with the smallest number of events in it. This notion of “smallest σ -algebra containing some events” arises frequently in probability theory.

Exercise 1a (4 points): Construct the σ -algebra generated by the following subsets of Ω_3 :

$$A_1 = \{HHH, HHT\} , \quad A_2 = \{HHT, HTH, TTH\} . \quad (3)$$

This is an example where the subsets that generate the σ -algebra have non-empty intersections amongst themselves. ■

An **atomic event** or simply an **atom** of a σ -algebra is an event in the σ -algebra which does not contain any other event in the σ -algebra except itself or the empty set $\emptyset$. Suppose a σ -algebra is generated by a finite number of subsets. For such a σ -algebra, its atoms have the following properties:

1. The intersection of any two atoms is empty.
2. The union of all the atoms is the sample space itself.
3. The collection of all atoms generates the σ -algebra.

Caution: There are σ -algebras which do not have any atomic events in them, or even if there are atoms, they may not generate the σ -algebra. However, we will not consider them in this course.

Note that a σ -algebra may not have been specified as being generated by atoms but by some other collection of a finite number of subsets. However, once we obtain the atoms of the σ -algebra, we can show that these also generate the same σ -algebra. For instance, let us look at the σ -algebra generated by the subsets $C = \{HTT, HTH\}$ and $D = \{HTH, HHH\}$ of Ω_3 . The σ -algebra is obtained by taking all possible unions, intersections and complements:

$$\begin{aligned}
C &= \{HTT, HTH\} , \quad D = \{HTH, HHH\} , \\
C \cup D &= \{HHH, HTH, HTT\} , \quad C \cap D = \{HTH\} , \\
C^c &= \{HHH, HHT, THH, THT, TTH, TTT\} , \\
D^c &= \{HHT, HTT, THH, THT, TTH, TTT\} , \\
(C \cap D)^c &= C^c \cup D^c = \{HHH, HHT, HTT, THH, THT, TTH, TTT\} , \\
(C \cup D)^c &= C^c \cap D^c = \{HHT, THH, THT, TTH, TTT\} , \\
C \cup D^c &= \{HHT, HTH, HTT, THH, THT, TTH, TTT\} , \\
D \cup C^c &= \{HHH, HHT, HTH, THH, THT, TTH, TTT\} , \\
C \cap D^c &= \{HTT\} , \quad D \cap C^c = \{HHH\} . \\
C \cup C^c &= D \cup D^c = \Omega_3 , \quad C \cap C^c = D \cap D^c = \emptyset .
\end{aligned} \tag{4}$$

The atoms of the above σ -algebra are

$$\begin{aligned}
C \cap D &= \{HTH\} , \quad C \cap D^c = \{HTT\} , \quad D \cap C^c = \{HHH\} , \\
C^c \cap D^c &= \{HHT, THH, THT, TTH, TTT\} .
\end{aligned} \tag{5}$$

It is easy to verify each of the three properties mentioned above for these atoms.

Exercise 1b (2 points): Describe the atomic subsets of the σ -algebra $\sigma(A_1, A_2)$ obtained in Exercise 1b.

Note that A_1, A_2 are not atomic subsets since they have a non-empty intersection, and also their union is not the sample space. ■

We now describe the σ -algebra generated by a random variable $\mathbb{X}$ which takes a finite set of values. The set of all values taken by a random variable $\mathbb{X}$ is called the *range* of $\mathbb{X}$ and is denoted $\text{Ran}(\mathbb{X})$ or $\text{Range}(\mathbb{X})$.

Consider subsets of sample space (events) of the form $\{\mathbb{X} = a\}$ where a is one of the possible values in $\text{Range}(\mathbb{X})$. For instance, when the total number of coin tosses is $N = 4$, the event $\{\mathbb{M}_2 = 2\}$ consists of the sequences $\{HHHH, HHHT, HHTH, HHTT\}$, i.e., all sequences starting with two heads. Using the collection of all such subsets, form all possible unions, intersections and complements. The resulting collection of subsets is closed under unions, intersections and complements and is called the σ -algebra generated by the random variable $\mathbb{X}$ and is denoted $\sigma(\mathbb{X})$.

Similarly, the σ -algebra generated by a collection of random variables $\mathbb{X}_1, \mathbb{X}_2, \dots, \mathbb{X}_n$ is obtained as follows. Consider all subsets of the form $\{\mathbb{X}_1 = a_1\}, \{\mathbb{X}_2 = a_2\}, \dots, \{\mathbb{X}_n = a_n\}$ where a_1 runs through the range of $\mathbb{X}_1$, a_2 runs through the range of $\mathbb{X}_2$ and so on. Form all possible unions, intersections and complements of all these subsets: this is the σ -algebra generated by $\mathbb{X}_1, \mathbb{X}_2, \dots, \mathbb{X}_n$ and is denoted $\sigma(\mathbb{X}_1, \mathbb{X}_2, \dots, \mathbb{X}_n)$.

Exercise 1c (6 points): Compute the σ -algebras $\sigma(\mathbb{M}_1)$ and $\sigma(\mathbb{M}_2)$ generated by the random variables $\mathbb{M}_1$ and $\mathbb{M}_2$ respectively. You must do this separately when $N = 3$ and when $N = 4$. ■

Exercise 1d (6 points): Let $\mathcal{A}_n$ be the σ -algebra generated by the random variables $\mathbb{M}_1, \mathbb{M}_2, \dots, \mathbb{M}_n$. That is, $\mathcal{A}_n = \sigma(\mathbb{M}_1, \mathbb{M}_2, \dots, \mathbb{M}_n)$. Compute $\mathcal{A}_1 = \sigma(\mathbb{M}_1)$, $\mathcal{A}_2 = \sigma(\mathbb{M}_1, \mathbb{M}_2)$, $\mathcal{A}_3 = \sigma(\mathbb{M}_1, \mathbb{M}_2, \mathbb{M}_3)$. For this question, you can take the total number of coin tosses to be $N = 3$. ■

We next define the notion of a σ -subalgebra and that of a filtration of σ -algebras. Given a σ -algebra $\mathcal{F}$, another σ -algebra $\mathcal{G}$ is said to be a σ -subalgebra of $\mathcal{F}$ if every event in $\mathcal{G}$ is also contained in $\mathcal{F}$. This is expressed as $\mathcal{G} \subseteq \mathcal{F}$.

A filtration of σ -algebras is a sequence of σ -algebras $\mathcal{G}_0, \mathcal{G}_1, \dots, \mathcal{G}_n$ such that each σ -algebra in the sequence is a σ -subalgebra of the next one in the sequence. That is, one has the following chain of σ -algebras $\mathcal{G}_0 \subseteq \mathcal{G}_1 \subseteq \dots \subseteq \mathcal{G}_n$.

Exercise 1e (2 points): Let us define $\mathcal{A}_0$ to be the trivial σ -algebra $\{\emptyset, \Omega\}$. Also recall the σ -algebras $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$ obtained in the previous exercise. Show that the sequence of

σ -algebras $\{\mathcal{A}_n\}_{0 \leq n \leq 3}$ is a filtration. That is, show that $\mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \mathcal{A}_2 \subseteq \mathcal{A}_3$. ■

Note: We frequently drop the extra qualification that $0 \leq n \leq N$ in the subscript of the filtration and simply write $\{\mathcal{A}_n\}$ for the filtration. An $\mathcal{A}_n$ without the enclosing braces will be used to refer the n^{th} σ -algebra in the filtration.

Let us now look at conditional expectations. These are defined based on conditional probabilities. Suppose we have events A and B and we want to compute the probability that A occurs given that B has occurred. This means we have to restrict our attention to that part of the sample space in which B has occurred. Recall that an event such as A or B correspond to certain subsets of the sample space. For instance, if we say B is the event that the first two coin tosses are H and T respectively, we have to restrict ourselves to the subset $B = \{\underline{\omega} \in \Omega_N \mid \omega_1 = H, \omega_2 = T\}$. Similarly, an example of A would be the event that the first three coin tosses are HTH .

Once we restrict ourselves to the subset B , we then look for occurrences of the event A within the subset B rather than the entire sample space. As discussed in class, this involves the joint occurrences of the event A and B which is conveyed by the intersection $A \cap B$ which is also sometimes written AB . The probability of the occurrence of A given B is then

$$\mathbb{P}\{A \mid B\} = \frac{\mathbb{P}\{AB\}}{\mathbb{P}\{B\}}. \quad (6)$$

Let us consider the situation where the number of coin tosses in a sequence is $N = 3$ so that we are dealing with the sample space of sequences of 3 coin tosses. When A is the event that the three coin tosses are HTH and B is the event that the first two coin tosses are HT , we have $A = \{HTH\}$, $B = \{HTH, HTT\}$, and $AB = \{HTH\}$. We have $\mathbb{P}\{B\} = p^2q + pq^2 = pq$ and $\mathbb{P}\{AB\} = p^2q$. Thus, we have

$$\mathbb{P}\{A \mid B\} = \frac{\mathbb{P}\{AB\}}{\mathbb{P}\{B\}} = \frac{p^2q}{pq} = p. \quad (7)$$

On the contrary, if A is the event that the first three coin tosses are TTH and B is the event that the first two coin tosses are HT , these events cannot jointly occur. So, $\mathbb{P}\{AB\} = 0$ in this case since $AB = \emptyset$. This gives $\mathbb{P}\{A \mid B\} = 0$.

In other words, one has to recompute the probability of occurrence of an event A given new information that the event B has occurred. This new probability is the conditional probability $\mathbb{P}\{A \mid B\}$ and is typically different from the original probability of occurrence of A which is simply $\mathbb{P}\{A\}$.

Another example which will be useful in defining conditional expectations is when we take the event A to be a singleton set, i.e., a set that contains a single element of the sample space $A = \{\underline{\omega}\}$. The original probability of a sequence with n_H heads and n_T tails is $p^{n_H}q^{n_T}$. Let

us again take B to be some event in the sample space (e.g., the subset of sequences starting with HT). If the sequence $\underline{\omega}$ is in B (that is, it starts with HT), then $AB = \{\underline{\omega}\}$. If $\underline{\omega}$ is not in B , then $AB = \emptyset$. Thus, the joint probability of $A = \{\underline{\omega}\}$ and B occurring is

$$\mathbb{P}\{\{\underline{\omega}\} \cap B\} = \begin{cases} \mathbb{P}\{\underline{\omega}\} & \text{if } \underline{\omega} \in B, \\ 0 & \text{if } \underline{\omega} \notin B. \end{cases} \quad (8)$$

This then gives the conditional probability

$$\mathbb{P}\{\underline{\omega} \mid B\} = \begin{cases} \frac{\mathbb{P}\{\underline{\omega}\}}{\mathbb{P}\{B\}} & \text{if } \underline{\omega} \in B, \\ 0 & \text{if } \underline{\omega} \notin B. \end{cases} \quad (9)$$

For example, in the case of B being the event where the first two coin tosses are HT , we have shown earlier that $\mathbb{P}\{B\} = pq$. We also know $\mathbb{P}\{\underline{\omega}\} = p^{n_H(\underline{\omega})}q^{n_T(\underline{\omega})}$ where $n_H(\underline{\omega})$ and $n_T(\underline{\omega})$ are the number of heads and tails in the sequence $\underline{\omega}$. This gives

$$\mathbb{P}\{\underline{\omega} \mid B\} = \begin{cases} p^{n_H(\underline{\omega})-1}q^{n_T(\underline{\omega})-1} & \text{if } \underline{\omega} \in B, \\ 0 & \text{if } \underline{\omega} \notin B. \end{cases} \quad (10)$$

Again, we see that the probability of occurrence of a sequence in sample space is revised if we incorporate the information that the first two coin tosses are HT .

The expectation value of a random variable $\mathbb{Y}$ given the knowledge that an event $B \subseteq \Omega_N$ has occurred is defined as

$$\mathbb{E}[\mathbb{Y} \mid B] = \sum_{\underline{\omega} \in \Omega_N} \mathbb{Y}(\underline{\omega}) \mathbb{P}\{\underline{\omega} \mid B\} = \sum_{\underline{\omega} \in B} \mathbb{Y}(\underline{\omega}) \mathbb{P}\{\underline{\omega} \mid B\}. \quad (11)$$

The main change compared to the formula for an ordinary expectation value is that the probability associated to each such element $\underline{\omega}$ is $\mathbb{P}\{\underline{\omega} \mid B\}$ and is typically different from the original probability $\mathbb{P}\{\underline{\omega}\}$. Note that the summation over all elements of the sample space can now be restricted to the subset B since the conditional probabilities for $\underline{\omega} \notin B$ are zero and hence do not contribute to the sum.

It is clear from the above formula that the conditional expectation value is fixed once the event B is specified. If the event B is changed to another event, then the expectation value may also change.

Thus, the conditional expectation value $\mathbb{E}[\mathbb{Y} \mid B]$ is a function of the event B . Just like conditional probability is a revised probability of the occurrence of an event, the conditional expectation $\mathbb{E}[\mathbb{Y} \mid B]$ is a revised expectation value of the random variable $\mathbb{Y}$ given that the event B has occurred.

Exercise 1f (2 points): Let us go back to the random walk. Let the event B be the subset of the sample space Ω_N consisting of sequences $\underline{\omega}$ with $\mathbb{M}_3 = 1$. That is,

$$B = \{\underline{\omega} \in \Omega_N \mid \mathbb{M}_3(\underline{\omega}) = 1\}, \quad (12)$$

also written in short as $B = \{\mathbb{M}_3 = 1\}$. Compute the conditional expectation value

$$\mathbb{E}[\mathbb{M}_4 \mid B] \quad \left(\text{also written as } \mathbb{E}[\mathbb{M}_4 \mid \mathbb{M}_3 = 1] \right), \quad (13)$$

using the formula (11). You should compute all the required probabilities that appear in the formula (11). For the purposes of this question, you can take the number of tosses to be $N = 4$ and borrow any required results from previous exercises. ■

As discussed above, the conditional expectation $\mathbb{E}[\mathbb{Y} \mid B]$ is a function of the event B . However, we know that information present in specifying one event B in sample space can be expressed in terms of the σ -algebra generated by the event B via complements, unions and intersections. This is in fact the σ -algebra

$$\sigma(B) = \{B, B^c, \emptyset, \Omega\}, \quad (14)$$

and is also the smallest σ -algebra of any possible σ -algebra that contains the event B . In other words, any other σ -algebra that contains B must contain more subsets than $\sigma(B)$.

The reason to consider this extra layer of abstraction is that, if one knows the probability $\mathbb{P}\{B\}$ of the event B occurring, one also immediately knows the probability $\mathbb{P}\{B^c\}$ of the event B^c occurring ($\mathbb{P}\{B^c\} = 1 - \mathbb{P}\{B\}$). Further, we know trivially that $B \cap B^c = \emptyset$ is the null event with $\mathbb{P}\{\emptyset\} = 0$, i.e., no event occurs, and $B \cup B^c = \Omega$ with $\mathbb{P}\{\Omega\} = 1$, i.e., some event occurs. Together, this σ -algebra completely summarizes the new information that *can* be specified once the event B is specified, along with the probabilities – either we say B occurs, or B^c occurs, or $B \cap B^c = \emptyset$ occurs, which is the same as the information that no sequence in the sample space has occurred, or finally $B \cup B^c = \Omega$ occurs, which is the same as having no prior information about any specific event occurring.

The expectation value of some random variable $\mathbb{Y}$ can now be computed for every event in $\sigma(B)$ and these expectation values give us a better understanding as to how the expected value of $\mathbb{Y}$ is revised given the information in $\sigma(B)$.

Exercise 1g (4 points): One can now compute the conditional expectation value of a random variable $\mathbb{Y}$ given each of the subsets in the σ -algebra $\sigma(B)$. Take $\mathbb{Y} = \mathbb{M}_4$, $B = \{\mathbb{M}_3 = 1\}$ as in the previous exercise and compute

$$\mathbb{E}[\mathbb{M}_4 \mid B^c], \quad \mathbb{E}[\mathbb{M}_4 \mid \emptyset], \quad \mathbb{E}[\mathbb{M}_4 \mid \Omega]. \quad (15)$$

For this question, you can take $N = 4$. ■

Next, we define the conditional expectation random variable $\mathbb{E}[\mathbb{X} \mid \mathcal{F}]$ which involves a random variable $\mathbb{X}$ and information given in terms of a σ -algebra $\mathcal{F}$. The definition we give here holds only for a σ -algebra which is generated by its atoms. (As mentioned earlier, not every σ -algebra has atoms, or even if it has atoms, may not be generated by them.)

Since the atoms of the σ -algebra $\mathcal{F}$ are all disjoint, it can be shown that any given element $\underline{\omega}$ of sample space belongs to one and only one atom. First, since the union of all atoms is the sample space, every element of sample space must belong to at least one atom. If a given element $\underline{\omega}$ belongs to more than one atom, then the two atoms have a common element $\underline{\omega}$ which means that they are no longer disjoint. Hence, every element $\underline{\omega}$ of sample space belongs to one and only atom of the σ -algebra $\mathcal{F}$. Note that this is true for any σ -algebra generated by its atoms.

This fact allows us to define the conditional expectation random variable $\mathbb{E}[\mathbb{X} \mid \mathcal{F}]$ by giving its value on each element of sample space. Let us call the unique atom containing a given element $\underline{\omega}$ as A^ω . Then, the value of $\mathbb{E}[\mathbb{X} \mid \mathcal{F}]$ on $\underline{\omega}$ is given by the conditional expectation value $\mathbb{E}[\mathbb{X} \mid A^\omega]$:

$$\mathbb{E}[\mathbb{X} \mid \mathcal{F}](\underline{\omega}) = \mathbb{E}[\mathbb{X} \mid A^\omega] . \quad (16)$$

Note that the notation on the left and right hand sides are very similar. On the left hand side, the conditioning is on a full σ -algebra and this conditional expectation is always interpreted as a random variable, i.e., a *function* on sample space. On the right hand side, the conditioning is on a particular atomic event A^ω and this is always a conditional expectation value, i.e., a *number*.

For instance, suppose B is an event, e.g., the event consisting of all coin toss sequences starting with HT . The σ -algebra generated by B is $\{B, B^c, \emptyset, \Omega\}$. Clearly, the events B and B^c are the atoms of $\sigma(B)$. Thus, the value of the random variable $\mathbb{E}[\mathbb{Y} \mid \sigma(B)]$ on the elements $\underline{\omega} \in \Omega$ are given by

$$\mathbb{E}[\mathbb{Y} \mid \sigma(B)](\underline{\omega}) = \begin{cases} \mathbb{E}[\mathbb{Y} \mid B] & \text{if } \underline{\omega} \in B , \\ \mathbb{E}[\mathbb{Y} \mid B^c] & \text{if } \underline{\omega} \in B^c . \end{cases} \quad (17)$$

Exercise 1h (4 points): Compute the random variables $\mathbb{E}[\mathbb{M}_3 \mid \sigma(\mathbb{M}_2)]$ and $\mathbb{E}[\mathbb{M}_3 \mid \mathcal{A}_2]$ where $\mathcal{A}_2 = \sigma(\mathbb{M}_1, \mathbb{M}_2)$. How are these answers different? For this exercise, you can take $N = 3$ and use the σ -algebras for this case. ■

OPTIONAL EXERCISES FOR MORE PRACTICE (WILL NOT BE GRADED):

Exercise 1i: Compute all the random variables $\mathbb{E}[\mathbb{M}_m \mid \sigma(\mathbb{M}_n)]$ for all $1 \leq m, n \leq N = 3$. Do you observe a pattern?

Exercise 1j: Compute all the random variables $\mathbb{E}[\mathbb{M}_m \mid \mathcal{A}_n]$ for all $1 \leq m, n \leq N = 3$. How are the σ -algebras $\mathcal{A}_n$ related to the σ -algebras $\mathcal{F}_n$ determined by the first n coin tosses that we discussed in class?